human

Find all tall buildings in Exeter that are over 18m tall or 7 floors high

host_agent

Tool generate_metadata_for_all_artifacts with args : {} Tool generate_metadata_for_all_artifacts returned result : {} Tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to find all tall buildings in Exeter that are over 18m tall or 7 floors high. Provide the general steps to solve this geospatial query.'}

planning_agent

1. The general geographical area is Exeter, so there will be 1 entry for Exeter. 2. The entities to be searched are tall buildings, specifically those that are over 18m tall or have 7 floors or more. 3. The conditions to apply are that the buildings must meet either of the height or floor criteria.

output steps: ["Find Exeter 1 area", "Search for tall buildings in Exeter as many search results", "Apply condition: buildings over 18m tall or 7 floors high"]

host_agent

Tool send_message returned result : 1. The general geographical area is Exeter, so there will be 1 entry for Exeter. 2. The entities to be searched are tall buildings, specifically those that are over 18m tall or have 7 floors or more. 3. The conditions to apply are that the buildings must meet either of the height or floor criteria.

output steps: ["Find Exeter 1 area", "Search for tall buildings in Exeter as many search results", "Apply condition: buildings over 18m tall or 7 floors high"] Tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_polygon_data'} Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_polygon_data,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..,\n count of records fetched : 98.\n ',] Tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_polygon_data' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_polygon_data'."}

data_analysis_agent

Tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_polygon_data']} Tool generate_metadata_for_artifacts returned result : [...metadata...] Tool Code Executor

```
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for polygons specifically representing Exeter
    # Use name1_text for the name, and descriptiongroup/description for settlement/city
    filtered = gdf[
        (
            (gdf['name1_text'].str.lower() == 'exeter') &
            (
                (gdf['descriptiongroup'].str.lower().isin(['settlement', 'land name'])) |
                (gdf['description'].str.lower().isin(['city', 'town', 'suburban area', 'village', 'part of settlement']))
            )
        )
    ]
    summary = f"Filtered polygons for Exeter. Number of matching polygons: {len(filtered)}."
    artifact_name = 'filtered_exeter_polygon_data'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement/city/town/village/suburban area/pa
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered polygons for Exeter have been found. There is 1 matching polygon representing Exeter. The results are saved as 'filtered_exeter_polygon_data', which contains only the polygon(s) representing Exeter, filtered by name and relevant settlement/city/town/village/suburban area/part of settlement type.

named_area_agent

Tool send_message returned result : ["Filtered polygons for Exeter have been found. There is 1 matching polygon representing Exeter. The results are saved as 'filtered_exeter_polygon_data', which contains only the polygon(s) representing Exeter, filtered by name and relevant settlement/city/town/village/suburban area/part of settlement type.Addtionally some data artifacts have been generated with names ['filtered_exeter_polygon_data'] and \n descriptions ['GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement/city/town/village/suburban area/part of settlement type.']", []] The polygon for Exeter has been successfully filtered and saved. The relevant data is in the file named `filtered_exeter_polygon_data`, which contains only the polygon(s) representing Exeter. Tool send_message returned result : ["The polygon for Exeter has been successfully filtered and saved. The relevant data is in the file named `filtered_exeter_polygon_data`, which contains only the polygon(s) representing Exeter.Addtionally some data artifacts have been generated with names ['filtered_exeter_polygon_data'] and \n descriptions ['GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement/city/town/village/suburban area/part of settlement type.']", []] Tool send_message with args : {'target': 'buildings', 'task_description': "Search for buildings in the area defined by the artifact 'filtered_exeter_polygon_data' that are over 18m tall or have 7 or more floors."}

buildings_agent

Tool call_os_ngd with args : {'filters': None, 'bbox': 'filtered_exeter_polygon_data', 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_all_buildings_for_height_floor_filter'} Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : [\'buildings_exeter_all_buildings_for_height_floor_filter\', \'buildingline_exeter_all_buildings_for_height_floor_filter\', \'buildingpart_exeter_all_buildings_for_height_floor_filter\'],\n Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for buildings in the area defined by the artifact \'filtered_exeter_polygon_data\' that are over 18m tall or have 7 or more floors. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Search for buildings in the area defined by the artifact \'filtered_exeter_polygon_data\' that are over 18m tall or have 7 or more floors. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Search for buildings in the area defined by the artifact \'filtered_exeter_polygon_data\' that are over 18m tall or have 7 or more floors. using filters None."],\n counts of records fetched are : [69169, 69342, 85480].\n ', [, ,]] Tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "From the artifact 'buildings_exeter_all_buildings_for_height_floor_filter', filter and return only those buildings that are either over 18m tall (using the 'relativeroofbase' column for building height) or have 7 or more floors (using the 'building_numfloors' column). Save the filtered results as 'exeter_buildings_over_18m_or_7floors'."}

data_analysis_agent

Tool generate_metadata_for_artifacts with args : {'artifact_names': ['buildings_exeter_all_buildings_for_height_floor_filter']} Tool generate_metadata_for_artifacts returned result : [...metadata...] Tool Code Executor

```
def filter_exeter_buildings_over_18m_or_7floors(data: list):
    import pandas as pd
    import geopandas as gpd
    gdf = data[0]
    # Filter for buildings over 18m tall (height_relativeroofbase_m > 18) or with 7 or more floors (numberoffloors >= 7)
    filtered = gdf[(gdf['height_relativeroofbase_m'] > 18) | (gdf['numberoffloors'] >= 7)]
    summary = f"Filtered buildings in Exeter that are either over 18m tall or have 7 or more floors. Total: {len(filtered)} buildings."
    artifact_name = 'exeter_buildings_over_18m_or_7floors'
    artifact_description = "GeoDataFrame of Exeter buildings that are either over 18m tall (using 'height_relativeroofbase_m') or have 7 or more floors (using
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered buildings in Exeter that are either over 18m tall or have 7 or more floors. Total: 55 buildings.

An artifact has been generated with the name: exeter_buildings_over_18m_or_7floors.

Description: GeoDataFrame of Exeter buildings that are either over 18m tall (using 'height_relativeroofbase_m') or have 7 or more floors (using 'numberoffloors'). Includes all original columns and geometry. Tool send_message returned result : ['Filtered buildings in Exeter that are either over 18m tall or have 7 or more floors. Total: 55 buildings.\n\nAn artifact has been generated with the name: exeter_buildings_over_18m_or_7floors.\n\nDescription: GeoDataFrame of Exeter buildings that are either over 18m tall (using \'height_relativeroofbase_m\') or have 7 or more floors (using \'numberoffloors\'). Includes all original columns and geometry.Addtionally some data artifacts have been generated with names [\'exeter_buildings_over_18m_or_7floors\'] and \n descriptions ["GeoDataFrame of Exeter buildings that are either over 18m tall (using \'height_relativeroofbase_m\') or have 7 or more floors (using \'numberoffloors\'). Includes all original columns and geometry."]', []]

buildings_agent

Reasoning: - The query is about finding buildings in a specific area (defined by 'filtered_exeter_polygon_data') that are either over 18m tall or have 7 or more floors. - The Ordnance Survey NGD tool does not directly filter by height or number of floors, so I retrieved all buildings in the area. - I then used the data analysis agent to filter the results using the columns 'relativeroofbase' (for height) and 'building_numfloors' (for number of floors).

Results: - 55 buildings in Exeter meet the criteria of being either over 18m tall or having 7 or more floors. - The filtered results are saved in the artifact: exeter_buildings_over_18m_or_7floors.

Artifact: - exeter_buildings_over_18m_or_7floors: GeoDataFrame of Exeter buildings that are either over 18m tall (using 'height_relativeroofbase_m') or have 7 or more floors (using 'numberoffloors'). Includes all original columns and geometry.

If you need further analysis or details about these buildings, please let me know! Tool send_message returned result : ['Reasoning:\n- The query is about finding buildings in a specific area (defined by \'filtered_exeter_polygon_data\') that are either over 18m tall or have 7 or more floors.\n- The Ordnance Survey NGD tool does not directly filter by height or number of floors, so I retrieved all buildings in the area.\n- I then used the data analysis agent to filter the results using the columns \'relativeroofbase\' (for height) and \'building_numfloors\' (for number of floors).\n\nResults:\n- 55 buildings in Exeter meet the criteria of being either over 18m tall or having 7 or more floors.\n- The filtered results are saved in the artifact: exeter_buildings_over_18m_or_7floors.\n\nArtifact:\n- exeter_buildings_over_18m_or_7floors: GeoDataFrame of Exeter buildings that are either over 18m tall (using \'height_relativeroofbase_m\') or have 7 or more floors (using \'numberoffloors\'). Includes all original columns and geometry.\n\nIf you need further analysis or details about these buildings, please let me know!Addtionally some data artifacts have been generated with names [\'exeter_buildings_over_18m_or_7floors\'] and \n descriptions ["GeoDataFrame of Exeter buildings that are either over 18m tall (using \'height_relativeroofbase_m\') or have 7 or more floors (using \'numberoffloors\'). Includes all original columns and geometry."]', []] Tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot the buildings from the artifact 'exeter_buildings_over_18m_or_7floors' to show all tall buildings in Exeter that are over 18m tall or have 7 or more floors."}

plotting_agent

Tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_buildings_over_18m_or_7floors']} Tool generate_metadata_for_artifacts returned result : [...metadata...] Tool Code Executor

```python
def plot_tall_buildings_exeter(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Get the GeoDataFrame
    gdf = data[0].copy()

    # Ensure CRS is EPSG:4326 for folium
    if gdf.crs is not None and gdf.crs.to_string() != 'EPSG:4326':
        gdf = gdf.to_crs(epsg=4326)

    # Create a folium map centered on the mean location of the buildings
    centroid = gdf.geometry.centroid.unary_union.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=14)

    # Add each building polygon to the map
    for idx, row in gdf.iterrows():
        # Popup with building info
        popup_text = f"Description: {row['description']}<br>Height (rel roofbase): {row['height_relativeroofbase_m']} m<br>Floors: {row['numberoffloors']}"
        folium.GeoJson(
            mapping(row['geometry']),
            name=f"Building {idx}",
            style_function=lambda x: {'fillColor': '#ff7800', 'color': '#ff7800', 'weight': 2, 'fillOpacity': 0.5},
            tooltip=popup_text
        ).add_to(m)

    # Save map
    map_filename = 'exeter_tall_buildings_map.html'
    m.save(map_filename)

    # Prepare summary
    summary_rows = gdf[['description', 'height_relativeroofbase_m', 'numberoffloors']].head(5)
    summary = f"Plotted {len(gdf)} tall buildings in Exeter (over 18m or 7+ floors). First 5:\n" + summary_rows.to_string(index=False)
    artifact_name = 'exeter_tall_buildings_map'
    artifact_description = 'Folium map showing all buildings in Exeter over 18m tall or with 7 or more floors.'
    return [summary, artifact_name, artifact_description, map_filename]
```

Plotted 55 tall buildings in Exeter (over 18m or 7+ floors). First 5: description height_relativeroofbase_m numberoffloors Commercial Building 18.8 NaN Power Station 29.4 NaN Unknown Building 25.5 NaN Mixed Use Building 29.7 9.0 Multiple Residential Accommodation 18.1 5.0

Artifact name: exeter_tall_buildings_map

Artifact description: Folium map showing all buildings in Exeter over 18m tall or with 7 or more floors.

Artifact data: exeter_tall_buildings_map.html

You can view the interactive map in the provided HTML file. Tool send_message returned result : ["Plotted 55 tall buildings in Exeter (over 18m or 7+ floors). First 5:\n description height_relativeroofbase_m numberoffloors\n Commercial Building 18.8 NaN\n Power Station 29.4 NaN\n Unknown Building 25.5 NaN\n Mixed Use Building 29.7 9.0\nMultiple Residential Accommodation 18.1 5.0\n\nArtifact name: exeter_tall_buildings_map\n\nArtifact description: Folium map showing all buildings in Exeter over 18m tall or with 7 or more floors.\n\nArtifact data: exeter_tall_buildings_map.html\n\nYou can view the interactive map in the provided HTML file.Addtionally some data artifacts have been generated with names ['exeter_tall_buildings_map'] and \n descriptions ['Folium map showing all buildings in Exeter over 18m tall or with 7 or more floors.']", []]

host_agent

All tall buildings in Exeter that are over 18m tall or have 7 or more floors have been identified and mapped. Here’s how your query was solved:

1. The area polygon for Exeter was found.
2. All buildings within Exeter were searched, and those that are either over 18m tall or have 7 or more floors were filtered.
3. A total of 55 buildings in Exeter meet your criteria.
4. These buildings have been plotted on an interactive map (artifact: exeter_tall_buildings_map).

If you need details about specific buildings or want to view the map, let me know!